

Niloufar Dizangian

Personal Information

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Education

Oct. 2023 – Dec. 2026	**University of Lorraine**, Nancy, France PhD Candidate in Electronic Systems Doctoral Thesis: Advanced High-Gain and Fault-Tolerant Electronic Interfaces for Small-Scale Energy Harvesting Systems Institut Jean Lamour (IJL); Supervisor: Prof. Philippe Poure; Co-Supervisor: Prof. Slavisa Jovanovic Defense scheduled: 2 December 2026
Sept. 2016 – Feb. 2019	**Shahid Beheshti University**, Tehran, Iran M.Sc. in Electrical Engineering — Power Systems GPA: 16/20 (3.2/4)
Sept. 2012 – Jul. 2016	**Shahid Beheshti University**, Tehran, Iran B.Sc. in Electrical Engineering — Power Systems GPA: 18.22/20 (3.88/4)
Sept. 2008 – Jul. 2012	**Farzanegan High School (NODET)**, Tehran, Iran Diploma in Mathematics and Physics GPA: 19.62/20

Research Interests

- Power Electronics
- High-gain DC/DC converter topologies for energy harvesting
- Fault-tolerant converters
- Fault diagnosis and fault-tolerant control in power electronics
- Single-switch cascaded converter architectures
- Control system design for power electronic converters
- MPPT and output-voltage regulation of two-stage converters
- AI and Machine Learning for Power Electronics
- AI and Machine Learning in fault detection of DC/DC converters
- Energy Harvesting Systems
- Sustainable Electronic Systems

Experiences

Feb. 2026 – Aug. 2026	**Master's Thesis Co-Supervisor** (with Prof. Philippe Poure), University of Lorraine, Nancy, France Thesis on a redundant synchronous switching strategy for open-switch fault tolerance of cascaded Boost converters in PV systems. Guided the student on converter design, simulation, experimental validation, and scientific research.
Oct. 2024 – Sept. 2026	**Teaching Assistant**, University of Lorraine, Nancy, France Course: Microelectronic Design. Supervised laboratory sessions and mentored undergraduate and Master's students in electronic system design.
Oct. 2025 – Dec. 2025	**Visiting Researcher**, Durham University, Durham, United Kingdom Contributed to the development of fault-tolerant DC–DC converters for energy harvesting, focusing on fault detection and tolerance strategies.
Jun. 2025 – Dec. 2026	**Doctoral Students Manager, ORION Club Mind^2**, University of Lorraine, Nancy, France Hands-on projects in intelligent materials, nanotechnologies, and embedded systems; mentoring students via workshops, seminars, and international exchanges.
Jun. 2025 – Dec. 2026	**President, IEEE Student Branch**, University of Lorraine, Nancy, France Led technical events and IEEE student activities, including a Career Development Event (Oct. 2026) with the ORION Mind^2 club and the university careers service.
Oct. 2023 – Present	**Member, IEEE Women in Engineering (WIE)**, University of Lorraine, Nancy, France Active participation in initiatives promoting gender diversity and women's engagement in electrical engineering and power electronics.
Feb. 2019 – Jul. 2023	**Research Assistant**, University of Tehran Photovoltaic Lab, Tehran, Iran
Sept. 2018 – Jul. 2023	**Lecturer and Teaching Assistant**, University of Tehran, Tehran, Iran Courses: Power Electronics, Electrical Circuits.
Sept. 2010 – Jul. 2023	**Mathematics and Physics Tutor**, Farzanegan High School, Tehran, Iran Courses: Mathematics and Physics.

Professional Memberships

- IEEE Member
- President, IEEE Student Branch, University of Lorraine
- IEEE Power Electronics Society (PELS)
- IEEE Industrial Electronics Society (IES)
- IEEE Dielectrics and Electrical Insulation Society (DEIS)
- IEEE Young Professionals (YP)

Publications

[1] N. Dizangian, S. Jovanovic, and P. Poure, "Sustainable-Driven Methodology for Two-Stage Energy Management Circuits for IoT Nodes," in *Proceedings of the IEEE ICECS 2024*, Nancy, France. doi:10.1109/ICECS61496.2024.10848894

[2] H. Gholizadeh, N. Dizangian, P. Poure, and S. Jovanovic, "A High-Gain Single-Switch DC-DC Converter Based on Cascaded Boost," in *Proceedings of the IEEE EEEIC/ICPSEurope 2024*, Rome, Italy. doi:10.1109/EEEIC/ICPSEurope61470.2024.10751535

[3] N. Dizangian, S. Jovanovic, and P. Poure, "Modeling and Experimental Verification of a Single-Switch Quadratic Boost DC–DC Converter with High Voltage Gain for Energy Harvesting," *Energies*, vol. 18, no. 20, p. 5447, 2025. doi:10.3390/en18205447.

Talks and Conference Service

- Talk, *"Reliable Power Electronics for Energy Harvesting: High-Gain DC–DC Converters and Fault-Tolerant Control Strategies,"* IEEE France WIE *Women & Energy* session, IEEE ENERGYCON 2026, CNAM, Paris Oct. 2026
- Invited talk for Master's students (M2 EEA), *"Doing research after a Master's?"*, University of Lorraine Jan.

2026 and Nov. 2026

- Oral presentation, IEEE EEEIC/I&CPS Europe 2024, Rome, Italy Jun. 2024
- Student Organizing Committee, IEEE I2MTC 2026, Nancy, France May 2026
- Student Organizing Committee, IEEE ICECS 2024, Nancy, France Nov. 2024

Certificates

- IEEE Region 8 Student and Young Professionals Congress 2026
- MATLAB, Simulink, Simscape, Power Electronics Simulation (MathWorks), 2023
- Energy Harvesting (Sungkyunkwan Univ.), 2025
- Machine Learning Specialization (DeepLearning.AI & Stanford), 2025
- Statut National Étudiant-Entrepreneur (Pépite France), 2025–2026

Honors and Awards

- Best Pitch Prize, IEEE HIVE Design-Thinking Workshop, IEEE ENERGYCON 2026, Paris Oct. 2026
- Best Poster Award, IEEE Region 8 Student and Young Professionals Congress, Kraków, Poland Jul. 2026
- Selected Participant with IEEE Region 8 Travel Funding (registration, accommodation and meals), IEEE Region 8 Student and Young Professionals Congress, Kraków, Poland Jul. 2026
- Selected Visiting Researcher, Durham University, UK Oct. 2025 – Dec. 2025
- President, IEEE Student Branch at University of Lorraine Jun. 2025 – Present
- DrEAM Mobility Grant, Université de Lorraine (Internationalisation Commission) May 2025
- Ranked 3rd, 10th AFFDU Lorraine Doctoral Grant Feb. 2025
- Doctoral Fellowship, University of Lorraine Oct. 2023 – Sept. 2026
- 10th highest cumulative GPA among Electrical Engineering students (100+) Jun. 2019
- Honorary Admission for Graduate Study, Electrical Engineering Department Aug. 2016
- 2nd highest cumulative GPA among Electrical Engineering students (110+) Jun. 2016
- Top 1% in the Nationwide University Entry Exam (380,000+ participants) Aug. 2012
- Semifinalist, 20th Iranian National Olympiad in Physics, National Organization for Development of Exceptional Talents (Sampad) Mar. 2010
- Member, National Organization for Development of Exceptional Talents (NODET/SAMPAD) (less than 1% acceptance rate) Sept. 2008

Technical Skills

Converter Design	Topology synthesis, CCM/DCM analysis, component sizing, transient analysis
Control	MPPT, voltage regulation, duty-cycle and switching-frequency control, fault-tolerant reconfiguration
Hardware & Testing	Prototyping, oscilloscopes, voltage/current probes, DC supplies, electronic loads, data acquisition
Modeling & Simulation	MATLAB Simulink, Simscape, PLECS, LTspice
Programming	MATLAB, C/C++
CAD & Other Tools	Cadence Virtuoso, AutoCAD, LaTeX, MS Office
Project Management	Organization of IEEE events, conference committees, Master's thesis co-supervision

Languages

Persian	Native	
English	Advanced	
French	Intermediate	
IELTS	7/9	(Reading: 7.5, Listening: 7.5, Speaking: 7, Writing: 6)

Hobbies and Interests

Sports	Yoga, CrossFit, Physical Fitness
Other	Audiobooks, Podcasts, Puzzles, Travelling, Hiking

References

Prof. Philippe Poure	Professor, University of Lorraine	philippe.poure@univ-lorraine.fr
Prof. Slavisa Jovanovic	Professor, University of Lorraine	slavisa.jovanovic@univ-lorraine.fr
Prof. Ehsan Jamshidpour	Professor, University of Lorraine	ehsan.jamshidpour@univ-lorraine.fr
Prof. Mahmoud Shahbazi	Professor, Durham University	mahmoud.shahbazi@durham.ac.uk